CIVIL ENGINEERING					
IV B.Tech – I Semester					
Course Code	**EXPERIMENTAL STRESS ANALYSIS Professional Elective-IV**	L	T	P	C
23A01702c		3	0	0	3

Course Objectives:

The objectives of this course are to make the student to:

1. Understand the principles and advantages of experimental stress analysis.
2. Explain strain measurement techniques using strain gauges and their applications.
3. Analyze strain rosettes and apply non-destructive testing methods for concrete.
4. Understand the fundamental principles of photoelasticity and its applications.
5. Apply two-dimensional photoelasticity methods for stress analysisin materials.

Course Outcomes (COs):

Upon successful completion of the course, students will be able to:

1. Explain the principles and merits of experimental stress analysis.
2. Demonstrate strain measurement using various strain gauge techniques.
3. Analyze strain rosette data and evaluate concrete structures using NDT methods.
4. Apply the theory of photoelasticity to determine stress distributionsin materials.
5. Utilize two-dimensional photoelasticity techniques for experimental stress analysis.

CO – PO Articulation Matrix:-

Course Outcomes	**PO 1**	**PO 2**	**PO 3**	**PO 4**	**PO 5**	**PO 6**	**PO 7**	**PO 8**	**PO 9**	**PO 10**	**PO 11**	**PO 12**	**PSO1**	**PSO2**
CO -1	3	2	-	-	-	2	-	-	-	1	-	1	1	1
CO -2	3	2	2	2	2	3	-	-	-	-	-	1	2	2
CO -3	2	2	3	2	2	3	2	-	-	-	-	1	2	2
CO -4	3	2	2	2	2	3	2	-	-	-	-	1	2	2
CO -5	2	2	2	2	2	3	2	-	-	-	-	1	2	2

UNIT – I

PRINCIPLES of EXPERIMENTAL APPROACH

Merits of Experimental Analysis Introduction, Uses of Experimental Stress Analysis Advantages of Experimental Stress Analysis, Different Methods –Simplification of Problems

UNIT – II

STRAIN MEASUREMENT USING STRAIN GAUGES

Definition of Strain and Its Relation of Experimental Determinations Properties of Strain-Gauge Systems-Types of Strain Gauges –Mechanical, Acoustic and Optical Strain Gauges. Introduction to Electrical Strain Gauges - Inductance Strain Gauges – LVDT – Resistance Strain Gauges – Various Types –Gauge Factor – Materials of Adhesion Base

UNIT – III

STRAIN ROSSETTES and NON – DESTRUCTIVE TESTING of CONCRETE

Introduction – The Three Elements Rectangular Rosette – The Delta Rosette Corrections for Transverse Strain Gauge. Ultrasonic Pulse Velocity Method –Application to Concrete.

Hammer Test – Application to Concrete.

UNIT – IV		

THEORY of PHOTOELASTICITY

Introduction –Temporary Double Refraction – The Stress Optic Law –Effects of Stressed Modelin A Polar Scope for Various Arrangements – Fringe Sharpening. Brewster's Stress Optic Law

UNIT – V		

TWO DIMENSIONAL PHOTOELASTICITY

Introduction – Isochramic Fringe Patterns- Isoclinic Fringe Patterns Passage of Light Through Plane Polariscope and Circular Polariscope - Isoclinic Fringe Patterns – Compensation Techniques – Calibration Methods – Separation Methods – Scaling Model to Prototype Stresses – Materials for Photo – Elasticity Properties of Photoelastic Materials.

TEXT BOOKS:

1. Experimental stress analysis by J.W.Dally andW.F.Riley, College House Enterprises2005
2. Experimental stress analysis by Dr.SadhuSingh.khanna Publishers 4th edition

REFRENCE BOOKS:

1. Experimental Stress analysis by U.C.Jindal, Pearson Pubilications 2012 edition
2. Experimental Stress Analysis by L.S.Srinath, MC.Graw Hill Company Publishers

Online Learning Resources:

https://archive.nptel.ac.in/courses/112/106/112106068/#
https://archive.nptel.ac.in/courses/112/106/112106198/#